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Editor-in-Chief
PLOS Computational Biology

Dear Editor,

We are pleased to submit our manuscript entitled “*CardamomOT: a mechanistic optimal transport-based framework for gene regulatory network inference, trajectory reconstruction and generative modeling*” for consideration as a Research Article in *PLOS Computational Biology*.

This work builds upon our previously published method CARDAMOM (PLOS Computational Biology, 2023), which has, to date, received over 40 citations and has contributed to establishing mechanistic modeling as a viable approach for gene regulatory network (GRN) inference from single-cell data. While the present study shares the same underlying mechanistic assumptions, it should not be viewed as a simple extension or incremental update. Instead, it introduces a fundamentally new framework that significantly advances the original methodology.

In CARDAMOM, inference relied on temporal snapshots but only exploited the relative ordering of timepoints, under a quasi-stationary approximation of protein dynamics. In contrast, the method introduced here, CardamomOT, explicitly leverages real temporal information and reconstructs latent protein trajectories through a mechanistic optimal transport formulation. This allows us to overcome key limitations of the original approach, particularly regarding the handling of system dynamics. More specifically, our contributions are threefold:

- We introduce a novel framework that jointly infers GRNs and cellular trajectories, a problem that has only recently begun to be addressed in the literature, and rarely within a mechanistic modeling paradigm.
- We explicitly reconstruct latent protein trajectories while incorporating only minimal prior information, namely protein degradation rates, which improves both robustness and biological interpretability.
- We obtain a fully calibrated mechanistic generative model, enabling *in silico* prediction of the effects of unseen perturbations.

Because our methodological contributions are central to the study and the pipeline structure directly informs how results are evaluated, we chose to present the Methods section prior to the Results section.

We validate our approach on both simulated datasets and three experimental single-cell time series datasets. On simulated data, we demonstrate improved performance in both GRN inference and trajectory reconstruction compared to state-of-the-art methods. On experimental datasets, we further show that our model can successfully predict the effect of genetic perturbations, including two predictions that are consistent with previously reported experimental findings.

We believe that this work will be of strong interest to the readership of *PLOS Computational Biology*, as it addresses central challenges in single-cell data analysis by combining mechanistic modeling, optimal transport, and GRN inference into a unified and interpretable framework.

Sincerely,

Yann Maugé and Elias Ventre